

FULL ARTICLE

Rational combinations of indirubin and arylidene derivatives exhibit synergism in human non-small cell lung carcinoma cells

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Abstract

Rational combination of natural and synthetic derivatives to treat lung cancer has advantages of both efficacy and safety. Herein, the combination of indirubin-3-monoxime (I3M), a chemical derived from Chinese herbal medicine and FXY-1, a synthetic arylidene derivative, was tested for combined activity in lung cancer cells. A dose-dependent synergistic reduction in cell viability was recorded with the combinations in A549 and NCI-H460 cells. Combination treatments of I3M and FXY-1 showed antimetastatic effects in both cells. Cell cycle analysis revealed G₁ growth phase reduction with subsequent accumulation of sub G₀ contents. Annexin V assay revealed higher apoptotic cells with combinations compared to individual treatments. I3M + FXY-1 combination significantly decreased the antiapoptotic Bcl-2 protein and increased pro-apoptotic Bax protein levels. These results demonstrate efficacy of I3M + FXY-1 in lung cancer cells and suggest further preclinical research in animal models to develop it into a new form combination chemotherapeutic against lung cancer.

Practical applications

Current investigation will open new options in rational combinations of natural and synthetic compounds to treat cancer. The observed efficacy and safety of the combinations will add to the advantage of higher therapeutic window in formulating treatment regimens. The antimetastatic effects by the combinations provides promising efficacy in controlling the lung cancer progression. A detailed *in vivo* investigation is recommended to transform the combinations to novel chemotherapeutic options against lung cancer.

KEYWORDS

apoptosis, arylidene, combination chemotherapeutics, FXY-1, I3M, indirubin, lung cancer, metastasis, synergism

1 | INTRODUCTION

Lung cancer (LC) is one of the leading death-related types of cancer that affects the major global population (Freddie, 2018). Out of total, almost 80% of LC cases are non-small cell lung carcinoma (NSCLC), while the other part is small cell lung carcinoma (Durate & Paschoal,

2006). Chemotherapy remains as the prime choice for NSCLC patients to improve the clinical symptoms and quality of life. However, the major limitation of chemotherapy includes, but not limited to the dose toxicities, drug resistance, and side effects (Mateen, Tyagi, Agarwal, Singh, & Agarwal, 2010). Consequently, search for more effective chemotherapeutics with minimal side effects is on demand.

Indirubin is identified as the most active component of *Danggui Longhui Wan*, a plant-based preparation used in Chinese traditional medicine. Indirubin is effectively used against various types of leukemia (Xiao, Hao, Liu, & Qian, 2002). Indirubin derivatives induce apoptosis and decrease the proliferative activity of tumor cells in RK3E-ras cell-injected Sprague dawley rats (Kim et al., 2007). The synthetic derivative of indirubin, indirubin-3-monoxime (I3M) (Figure 1a), has a better solubility and increased drug scaffold than indirubin (Varela et al., 2008; Zhang et al., 2009). I3M induces antiproliferative activity and promotes apoptosis in renal cancer and laryngeal carcinoma cells (Kameswaran & Ramanibai, 2009; Perabo et al., 2009). The effect of I3M on the morphological features of lungs in normal, [B (a) P] induced and I3M-treated A/J mice is also reported (Ravichandran, Pal, & Ravichandran, 2010). On the other hand, indan-1-ones are reported to inhibit cancer proliferation and inverse the multidrug resistance by standard anticancer drugs (Kars et al., 2006). Structure of FXY-1 (IUPAC-2-arylidene-4,7-dimethyl indan-1-one) is shown in Figure 1b. FXY-1 was proven against breast adenocarcinoma cells and lung cancer cells (Prasanna & Harish, 2010). FXY-1 was reported to induce apoptosis in human lung carcinoma cells through inhibiting AKT-mitochondrial pathway (Rajagopalan et al., 2016).

Though rich background information on the anticancer properties of these individual compounds are available, their combined efficacy in lung cancer cells has not been reported. It is because of the high degree of overlap between different signaling pathways and the redundancy within the cell, agents with multiple mechanisms of action, particularly the use of combination treatments have become increasingly important. Therefore, the current study aims to evaluate the combination effects of I3M and FXY-1 which have different mechanisms of action against lung cancer cells. The study aims to test the combinations in A549 and NCI-H460 cells which possess most of the cancerous cellular properties and are routinely used lung cancer models in vitro (Castell, Teresa Donato, & Gómez-Lechóna, 2005).

2 | MATERIALS AND METHODS

All chemicals and reagents were from Sigma (St. Louis, MO). Lung cancer cells lines (A549, NCI-H460), normal noncancerous cell line (HFL-1 and HUVEC) were revived from our cell line repository was originally obtained from the American Type Culture Collection (ATCC). HTEpiC cell line was obtained from Science Cell

Laboratories Ltd. Lung cancer cells were grown in RPMI-1640 medium. HUVEC cells were grown in F12K medium, containing 0.1 mg/ml heparin, 0.05 mg/ml endothelial cell growth supplement. HTEpiC cells were maintained in Bronchial Epithelial Cell Medium (BEpiCM, Cat. No. 3211, Science Cell Research Laboratories, USA). HFL-1 cells were maintained in F12K complete growth Medium. All growth medium contained FBS (10%), penicillin (100 U/ml), and streptomycin (100 U/ml), the cells were incubated in 5% CO₂ incubator at 37°C for growing conditions. Assays were carried out at 70% confluency of cell growth. Annexin V kit was purchased from e-Bioscience, USA. Guava Cell Cycle reagent was from Millipore Corp., USA. Bax, Bcl2, β -actin antibodies were from Cell Signaling, USA.

2.1 | Cell proliferation assay

Proliferation assay was performed as described by Mosman (1983) with modifications. Approximately 5,000 cells/well in regular growth media were incubated overnight and 50 μ l of single or combination compound was added along with a DMSO blank. The cells were incubated at 37°C and 5% CO₂ for 72 hr, and then 5 mg/ml MTT in 25 μ l was added and incubated for 4 hr. After media aspiration, the components were dissolved in 250 μ l of DMSO and read at 560 nm for absorbance 640 nm was used as reference wavelength. After subtracting day 0 values, the percent cell proliferation inhibition was calculated and results were analyzed using Graph pad Prism 6.0 software (La Jolla, USA).

2.2 | Scratch wound assay

NCI-H460 and A549 cells were grown in a 6-well plate until confluent. Approximately 1-mm channel was cut with a sterile 200- μ l pipette tip and the growth medium was replaced with serum-free medium containing individual or combination drug concentrations along with 50 ng/ml Hepatocyte growth factor (HGF) at the final concentration. Plates were incubated overnight in 37°C, 5% CO₂ incubator. Cells were compared with controls under microscope and pictures were taken at 4X.

2.3 | Transendothelial cell migration assay

The assay was carried out with QCM™ Tumor Cell Transendothelial Migration Assay—Colorimetric kit from Millipore as per the manufacturer's instructions. Briefly, 1×10^5 HUVEC cells were grown on the migration inserts until confluence. About 1×10^5 of NCI-H460 or A549 were starved in serum free medium for 12 hr. These cancer cells were replaced with the medium of migration inserts containing HUVEC cells and were moved to new wells that contained serum-free media ± 25 ng/ml HGF (final concentration). After 12-hr incubation in CO₂ incubator, the medium was removed from the inserts and elution was performed as per kit directions. The elutes were read for the absorbance at 540–570 nm. The data were analyzed with GraphPad Prism 6.0 software. Percentage inhibition of cell migration with respect to control was presented.

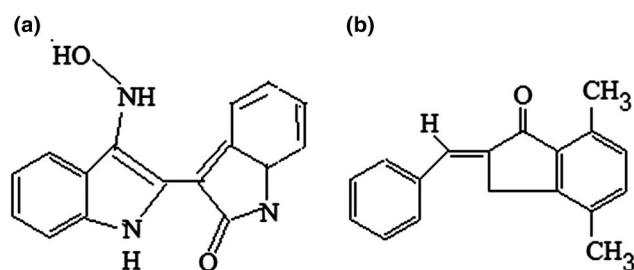


FIGURE 1 Chemical structure of (a) I3M and (b) FXY-1

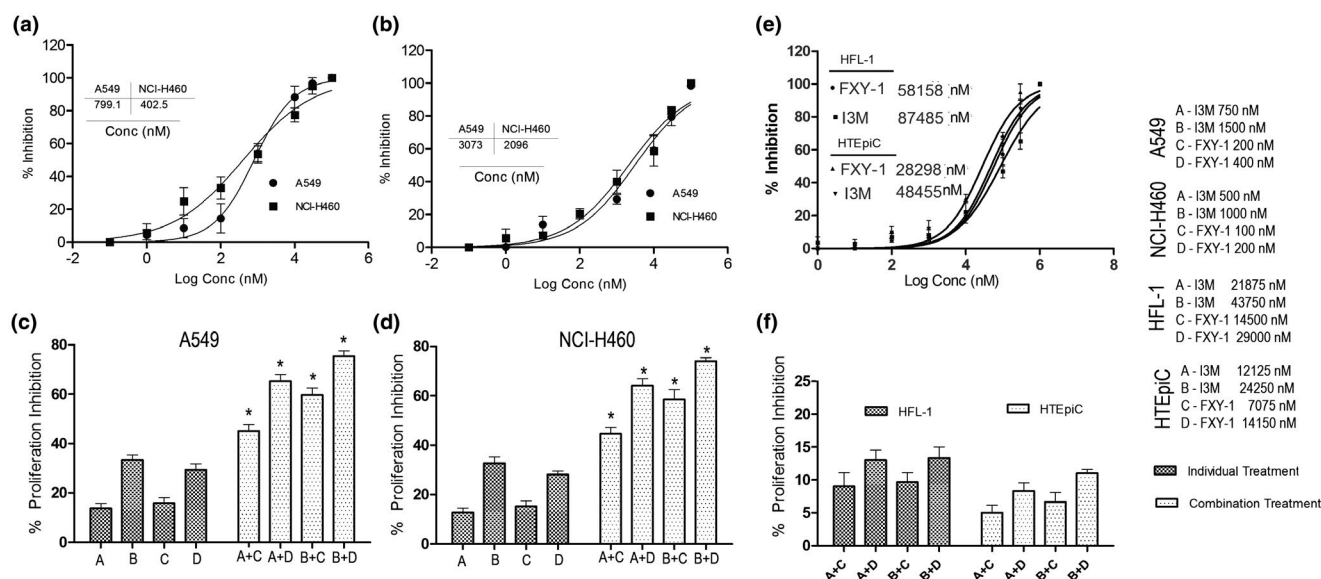


FIGURE 2 Antiproliferative profile of I3M and FXY-1 in lung cancer cells. (a) GI_{50} for FXY-1 in A549 and NCI-H460 cells. (b) GI_{50} for I3M in A549, NCI-H460 cells. Percentage proliferation inhibitions of I3M, FXY-1, and their combinations in (c) A549 and (d) NCI-H460 cells. (e) GI_{50} of FXY-1 and I3M in human lung fibroblast and tracheal epithelial cell lines. (f) Percentage proliferation inhibitions of combination in normal control cells. * $p \leq 0.05$ significant compared to control and individual treatments

2.4 | Cytokine estimations by enzyme-linked immunosorbent assay

Effects of I3M, FXY-1 and the combinations on the inflammatory mediator TNF- α release from A549 and NCI-H460 cells were analyzed by enzyme-linked immunosorbent assay. Cancer cells were plated in 6-well plates and were incubated with different concentrations of treatments for 1 hr, followed by 10 mg/ml LPS induction for 24 hr. the collected supernatants were stored in -80°C until analysis. Levels of TNF- α were assessed by kit as per the manufacturer's instructions. The color developed was measured for absorbance at 450 nm.

2.5 | Cell cycle analysis

Cells were seeded at a density of 0.5×10^6 cells per well in a 6-well plate and incubated with desired concentrations of single drugs or their combinations for 72 hr at 37°C in a 5% CO_2 incubator. The assay was carried out with Guava® Cell cycle reagent according to the manufacturer's instructions. Ten thousand events were acquired on a Guava easyCyte™ flow cytometer and the data analyzed with Express Pro Software from Millipore, USA. Percentage cell population at different stages of cell cycle was calculated.

2.6 | Annexin V assay

The assay was performed using Annexin V detection kit from e-Biosciences, USA, as per the manufacturer's instructions as follows. Briefly, 0.5×10^6 cells were grown in 6-well plates and treated with respective concentrations of FXY-1, I3M or their combinations followed by incubation in CO_2 incubator for 72 hr. Cells were harvested,

washed with kit buffer, and incubated with 0.25 $\mu\text{g/ml}$ Annexin V reagent for 15 min, washed again, and resuspended in kit buffer containing 0.5 $\mu\text{g/ml}$ propidium iodide. Ten thousand events were acquired on a guava easyCyte™ flow cytometer. Data analysis was carried out with InCyte software from Millipore, USA. Healthy cells were differentiated with different phases of apoptosis using a quadri plot graph. Data were analyzed using Graphpad Prism software—version 6.0 (La Jolla, USA).

2.7 | Western immunoblotting (pro- and antiapoptotic protein signaling)

Western blotting procedure was performed as described elsewhere (Rajagopalan et al., 2016), with minor modifications as follows. A549 and NCI-H460 cells were treated with either FXY-1, I3M or their combinations for 48 hr and lysed using lysis buffer. Total protein concentration was estimated by Coomassie Plus Protein Assay Reagent kit (Pierce; Rockford, IL, USA) and 20–40 μg of protein from the cell lysate was separated with SDS-PAGE followed by transferring to nitrocellulose membrane. After probing with respective primary antibodies, HRP—secondary antibodies were added, and the membrane was stripped after 30-min incubation. Bands were quantified using Image J (Ver. 1.46, NIH) and normalized to actin to β -actin (1:5,000).

2.8 | Statistical analysis

All experiments were performed in triplicates, and results expressed as the mean $\pm$ S.D. Statistical analyses were carried out with Graph pad Prism 6.0 (La Jolla, USA). IC_{50} or GI_{50} values were calculated using a nonlinear regression fit model with variable slope and plotted accordingly. Differences between two groups were analyzed using

the two-tailed Student's *t*-test and $p < 0.05$ (*/**) were considered statistically significant.

3 | RESULTS

3.1 | I3M and FXY-1 inhibited lung cancer cell proliferation as individuals and combinations in a dose-dependent manner

We examined the antiproliferative effects of I3M and FXY-1 in the lung cancer cells at different incubation check points like 24, 48, and 72 hr. I3M and FXY-1 clearly demonstrated antiproliferative effects against A549 and NCI-H 460 cells. FXY-1 showed GI_{50} (dose required for 50% growth inhibition of cell proliferation) values close to 0.8 μ M and 0.4 μ M, respectively, for A549 and NCI-H460 cells at 72 hr (Figure 2a). GI_{50} for I3M in A549 and NCI-H460 cells were close to 3 and 2 μ M, respectively (Figure 2b). To check out the combined effects of these candidates the GI_{25} and $GI_{12.5}$ doses were combined and treated to lung cancer cells. For A549 cells, $GI_{12.5}$ of I3M (750nM) was combined with $GI_{12.5}$ and GI_{25} of FXY-1 that is, 200 and 400 nM, respectively. GI_{25} of I3M (1500 nM) was combined with $GI_{12.5}$ and GI_{25} of FXY-1. Similarly, for NCI-H460 cells, the respective $GI_{12.5}$ of I3M was combined with

$GI_{12.5}$ and GI_{25} of FXY-1. As evident from Figure 2c,d, the combined drug effects were significantly more than the individual drugs alone or their mere individual effects added together. These observed combination effects were further checked for synergism by established formulas. Tables 1 and 2 show the synergistic effects of different combinations in both lung cancer cells. In addition, the individual compounds did not affect the viability of normal human tracheal epithelial cells and lung fibroblasts at higher fold concentrations suggesting a broader therapeutic window in vitro (Figure 2e). We also checked if the combination treatment had any effect in these normal cells. As observed in Figure 2f, the combinations had very little antiproliferative effects on the epithelial cells and lung fibroblast suggesting a safety therapeutic margin used for the combination treatments in lung cancer cells.

3.2 | I3M, FXY-1, and combinations exerted efficacy in preventing tumor metastasis of lung cancer cells

In order to check if there was any antimetastatic effect of the compounds and their combinations, we performed scratch wound assay and tumor cell trans endothelial migration analysis. Both the compounds as individual treatments or combinations were effective in preventing the tumor cells from healing of the scratch

TABLE 1 Effects of I3M and FXY-1 combination in A549 and NCI-H460 cells by observed to expected ratio

Cells	Treatment	Cell viability		Ratio ^b
		Observed %	Expected % ^a	
A549	DMSO control	100		
	I3M (750 nM)	86.32 $\pm$ 2.03		
	I3M (1,500 nM)	77.2 $\pm$ 2.02		
	FXY-1 (200 nM)	84.93 $\pm$ 2.16		
	FXY-1 (400 nM)	72.42 $\pm$ 2.43		
	I3M (750 nM) + FXY-1 (200 nM)	55.35 $\pm$ 2.72	73.31	1.32 [*]
	I3M (750 nM) + FXY-1 (400 nM)	33.83 $\pm$ 2.71	62.51	1.84 [*]
	I3M (1,500 nM) + FXY-1 (200 nM)	40.64 $\pm$ 2.90	65.56	1.61 [*]
	I3M (1,500 nM) + FXY-1 (400 nM)	26.81 $\pm$ 2.20	55.90	2.08 [*]
NCI-H460	DMSO control	100		
	I3M (500 nM)	87.34 $\pm$ 1.85		
	I3M (1,000 nM)	67.32 $\pm$ 2.60		
	FXY-1 (100 nM)	84.81 $\pm$ 2.24		
	FXY-1 (200 nM)	71.87 $\pm$ 1.89		
	I3M (500 nM) + FXY-1 (100 nM)	53.47 $\pm$ 2.72	74.07	1.38 [*]
	I3M (500 nM) + FXY-1 (200 nM)	35.88 $\pm$ 2.76	62.77	1.74 [*]
	I3M (1,000 nM) + FXY-1 (100 nM)	40.45 $\pm$ 3.96	57.09	1.41 [*]
	I3M (1,000 nM) + FXY-1 (200 nM)	24.92 $\pm$ 2.41	48.38	1.94 [*]

Note. Data are represented as mean $\pm$ SD of two independent experiments each performed in triplicate.

^aExpected value of I3M and FXY-1 combination = [(observed I3M treatment value)/(Control value)] $\times$ [(observed FXY-1 treatment value)/(Control value)] $\times$ (Control value). ^bRatio = (expected value/observed value). A ratio of >1 indicates a synergistic effect, and a ratio of <1 indicates a less than additive effect. *Significant synergism.

TABLE 2 Effects of I3M and FXY-1 combination in A549 and NCI-H460 cells by observed to mean ratio

	Treatment	Cell viability		Ratio ^b
		Observed %	Mean viability ^a	
A549	DMSO control	100		
	I3M (750 nM)	86.32 ± 2.03		
	I3M (1,500 nM)	77.2 ± 2.02		
	FXY-1 (200 nM)	84.93 ± 2.16		
	FXY-1 (400 nM)	72.42 ± 2.43		
	I3M (750 nM) + FXY-1 (200 nM)	55.35 ± 2.72	85.62	1.54 [*]
	I3M (750 nM) + FXY-1 (400 nM)	33.83 ± 2.71	79.37	2.34 [*]
	I3M (1,500 nM) + FXY-1 (200 nM)	40.64 ± 2.90	81.06	1.99 [*]
	I3M (1,500 nM) + FXY-1 (400 nM)	26.81 ± 2.20	74.81	2.79 [*]
NCI-H460	DMSO control	100		
	I3M (500 nM)	87.34 ± 1.85		
	I3M (1,000 nM)	67.32 ± 2.60		
	FXY-1 (100 nM)	84.81 ± 2.24		
	FXY-1 (200 nM)	71.87 ± 1.89		
	I3M (500 nM) + FXY-1 (100 nM)	53.47 ± 2.72	86.07	1.60 [*]
	I3M (500 nM) + FXY-1 (200 nM)	35.88 ± 2.76	79.62	2.21 [*]
	I3M (1,000 nM) + FXY-1 (100 nM)	40.45 ± 3.96	76.06	1.88 [*]
	I3M (1,000 nM) + FXY-1 (200 nM)	24.92 ± 2.41	69.59	2.79 [*]

Note. Data are represented as mean ± SD of two independent experiments each performed in triplicate.

^aMean cell viability of I3M and FXY-1. ^bRatio = (mean value/observed value). A ratio of >1 indicates a synergistic effect, and a ratio of <1 indicates a less than additive effect. *Significant synergism.

under influence of HGF (Figure 3a,b). This activity was very much observed to be increased in the combinations in comparison with separate treatment for both A549 and NCI-H460 cells (Figure 3a,b). Combination of 200 nM FXY-1 with 750 and 1,500 nM I3M inhibited 47.3% and 64.34% migration of A549 cells across HUVEC membrane, respectively (Figure 3c), while the percentage of inhibition for A549 cell migration was 54.29% for 750 nM I3M + 400 nM FXY-1 combination and 75.4% for 1,500 nM I3M + 400 nM combination. Similarly, 500 nM I3M + 100 nM FXY-1 inhibited the NCI-H 460 cell migration by 47.6% and 1,000 nM I3M + 100 nM FXY-1 combination inhibited the migration by 64.5% (Figure 3d). The inhibitory efficacy was found to be 57.5% for 500 nM I3M + 200 nM FXY-1 combination and 73% for 1,000 nM I3M + 200 nM FXY-1 combination when tested in NCI-H 460 cells (Figure 3d). Induction of A549 and NCI-H460 cells significantly induced TNF- α release as observed in Figure 3e,f. In A549 cells, 750 nM I3M and 200 nM FXY-1 combination reduced the TNF- α release to 370 pg/ml from 737.5 pg/ml as found in induction control (Figure 3e). Combining 400 nM FXY-1 to 750 nM I3M, further reduced the TNF- α concentration to 243 pg/ml in A549 cells (Figure 3e). At the highest combination concentrations of I3M + FXY-1, the TNF- α release was further reduced to 144 pg/ml which was close to the uninduced control (Figure 3e). 500 nM I3M + 100 nM FXY-1 combination in NCI-H 460 cells reduced the TNF- α level to 380 pg/ml from 880.3 pg/ml as found in the induction control (Figure 3f). The TNF- α release was 164 pg/ml when

treated with 1,000 nM I3M + 200 nM FXY-1 combination in NCI-H 460 cells (Figure 3f).

3.3 | I3M, FXY-1, and their combinations altered cell cycle progression of lung cancer cells

To investigate whether there is a role of I3M, FXY-1 and the combinations on cell cycle regulation, we analyzed the changes in cell cycle profiles by using flow cytometry. As shown in Figure 4, incubation of A549 and NCI-H 460 cells with I3M, FXY-1 or their combinations significantly increased the proportion of cells with a reduced DNA content (Sub G₀ region). Figure 4a,b shows the increase in M₄ or sub G₀ cells dose dependently with individual and combination treatments in all the two lung cancer cell types, which was further associated with a concordant decrease in the M₁ G₀/G₁ or growth phase of the cell cycle (Figure 4).

3.4 | Individual and combination treatments induced apoptosis in lung cancer cells by altering signaling proteins

Next, we analyzed the apoptosis inducing effects of individual agents and their combinations in the lung cancer cells by Annexin V assay. As observed in Figure 5, the individual treatments resulted in Annexin V binding to A549 and NCI-H 460 cells causing them to

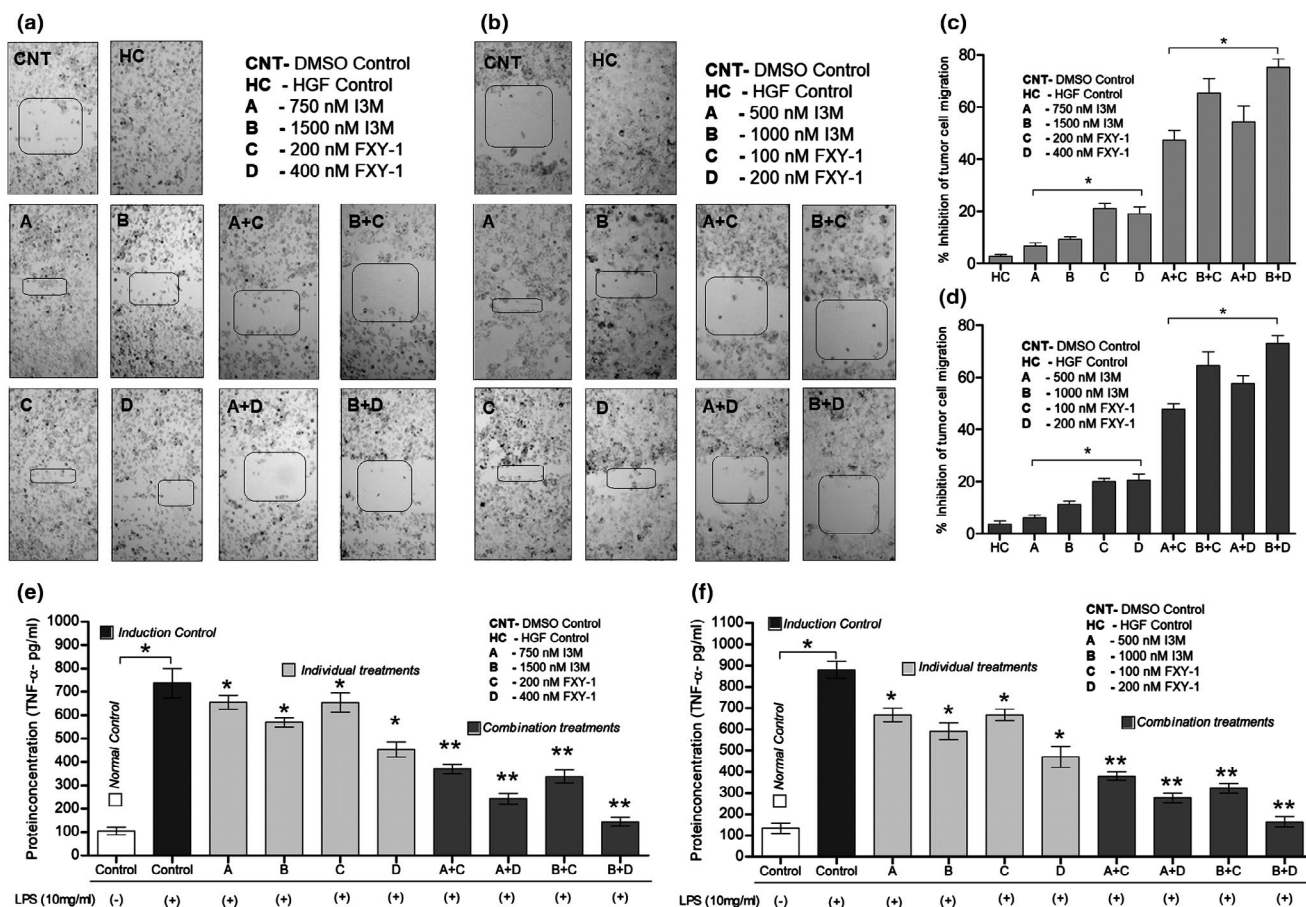


FIGURE 3 Effect of I3M, FXY-1, and their combinations in scratch wound healing assay for (a) A548 and (b) NCI-H 460 cells. Dose-dependent effects were observed in both the cells with increase in the concentrations of the individual components of combinations. I3M and FXY-1 inhibited the migration of (c) A549 and (d) NCI-H460 cells across the HUVEC membrane under influence of 25 ng/ml HGF which served as chemoattractant. The combinations had higher migration inhibitory effects compared to individual treatments which were found to be dose dependent (c, d). Results expressed as mean $\pm$ SD from three individual experiments in duplicates and * $p \leq 0.05$ was considered statistically significant. Inhibition of TNF- α release by (e) A549 and (f) NCI-H460 cells after individual or combination treatments. *significantly decreased compared to control, **significantly decreased compared to control and individual treatments ($p < 0.005$)

appear in the early and late phase apoptotic quadrants of the plots. The effect was prominently increased in the combination treated cells (Figure 5a,b). Together, the total apoptosis also remarkably increased in the combination for both lung cancer cell types (Figure 5a,b) which was in accordance with the preliminary observations of the antiproliferative assay. To substantiate the flow cytometric observations, we examined the effects of I3M, FXY-1, and their combinations for the apoptosis regulatory, Bcl-2 and Bax levels. Incubation of A459 and cells with individual compounds and their combinations, significantly decreased the antiapoptotic Bcl-2 protein and increased the pro-apoptotic Bax protein levels (Figure 6a,b). These observations were dose dependent as observed in other assays. When the protein bands were quantified and normalized to actin, a lenient regression of Bcl-2/Bax ratio was evident in treatment groups compared with the control (Figure 6). This effect increased in the combination treatments for both A549 and NCI-H460 cells with increase in the concentrations (Figure 6a,b).

4 | DISCUSSION

Safety and higher patient acceptance rate have been the prime focus in search of anticancer agents in recent times. Aforesaid objectives may be achieved either as individual agents or rational combinations of two or more chemicals together. The present study demonstrates the cytotoxicity indices of I3M, FXY-1, and their combinations as a measure of percentage cell viability calculated by MTT assay in lung cancer cells. The GI_{50} s of both drugs were in accordance to the previously reported literatures. Further, the present study exhibits more than individual additive cytotoxic effect in the tested combinations as observed in Figure 2. To rationalize the observed combination, two different formulas were used to study synergism as described elsewhere (Prasanna, Ashraf, & Essam, 2016). Table 1 uses a formula to calculate the expected rational outcome of the combination based on the individual treatment effects. The ratio of actual observed value obtained by combination to expected value

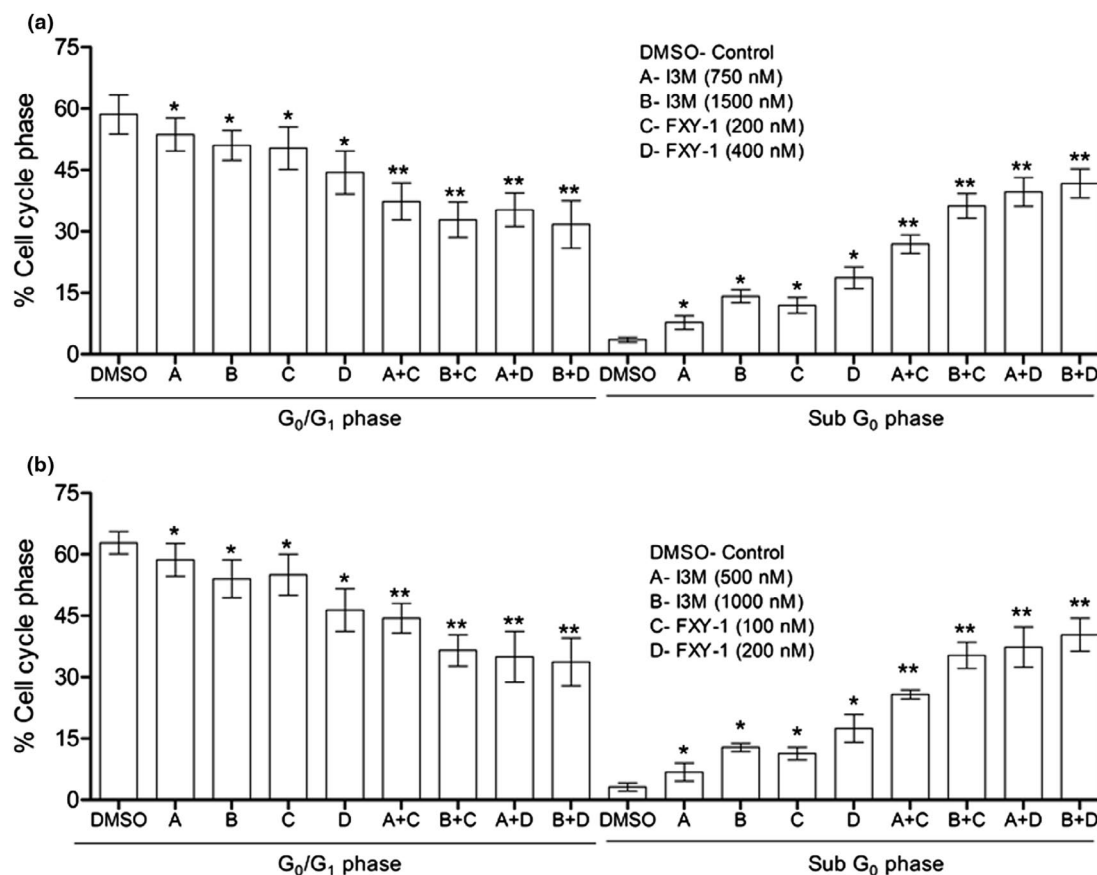


FIGURE 4 Flow cytometric analysis of cell cycle changes after separate and combinations treatments of I3M and FXY-1 in (a) A549 and (b) NCI-H460 cells for 72 hr. Representative histograms from several repeats of the experiment shows percentage cell population in different cell cycle stages with respect to untreated control. Results expressed as mean $\pm$ SD. Significantly ($p \leq 0.05$) increased compared to either *control or ** control and individual treatments

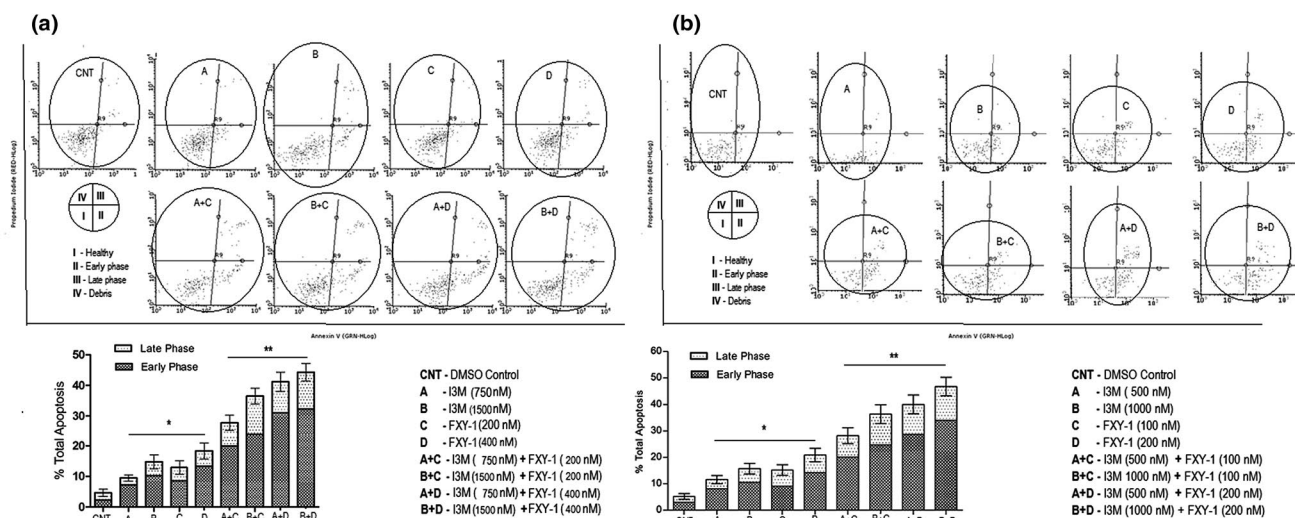


FIGURE 5 Representative figures from the flow cytometric enumeration for the early, late phase apoptotic cells in (a) A549 and (b) NCI-H460 cells determined by Annexin V staining. Bar chart shows a dose-dependent increase of total apoptotic cells in both cell lines when treated with I3M, FXY-1, and their combinations. Results are expressed as mean $\pm$ SD. Significant $p \leq 0.05$ compared to *control, **control, and individual treatments

is calculated by this method. A value lesser than 1 shows an additive effect, whereas greater than 1 is indicative of synergism. We used similar concentration combinations of I3M + FXY-1 ($GI_{12.5} + GI_{12.5}/GI_{12.5}$)

and varied concentration combinations ($GI_{12.5} + GI_{25}/GI_{25} + GI_{12.5}$) to check the effects. With this formula, the order of synergism in the combinations for both lung cancer cell lines were $GI_{12.5}$

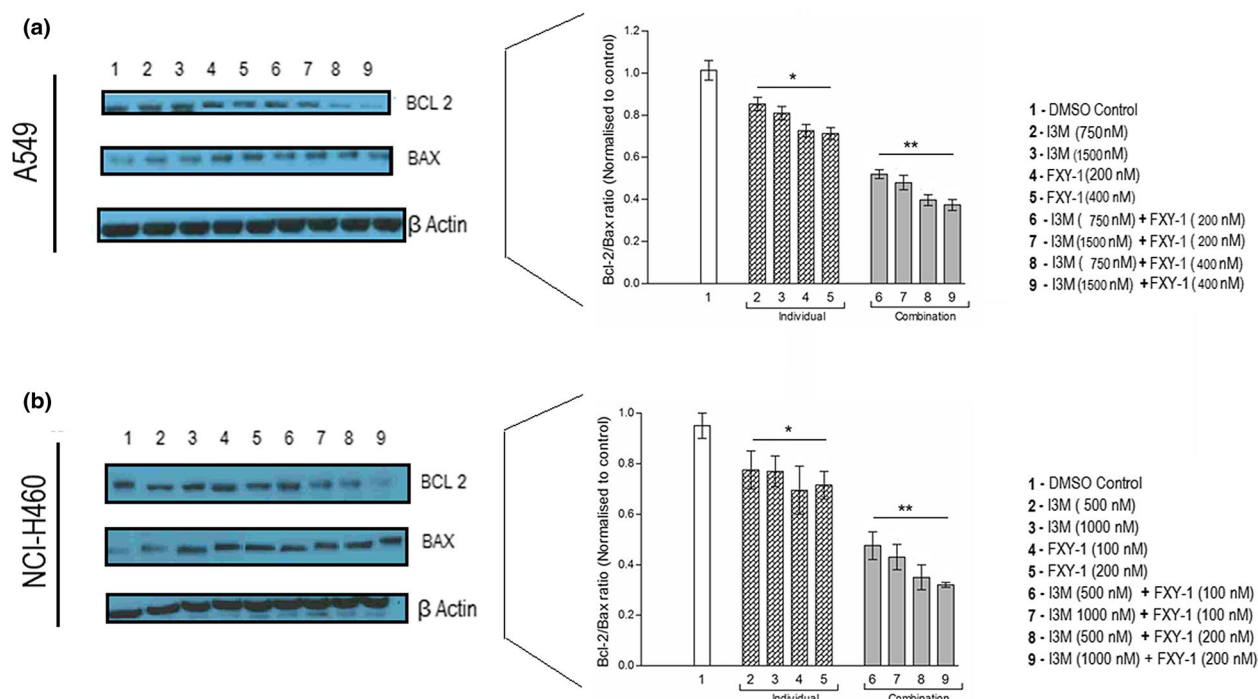


FIGURE 6 The effect of I3M, FXY-1, and their combinations on Bcl-2/Bax ratio in (a) A549 cells, (b) NCI-H460 cells. Representation blots are from three individual experiments performed in duplicates. β actin was used as loading control. Error bar represents SD of three independent experiments. *significantly decreased compared to control ($p < 0.005$), **significantly decreased compared to control and individual treatments ($p < 0.005$)

$I3M + GI_{12.5} FXY-1 < [GI_{25} I3M + GI_{12.5} FXY-1] < [GI_{12.5} I3M + GI_{25} FXY-1] < [GI_{25} I3M + GI_{25} FXY-1]$.

Table 2 uses the mean values of individual treatments. Here, a ratio of observed effect to mean value of individual treatments is calculated. Again, value lesser than 1 shows an additive effect, whereas greater than 1 is indicative of synergism. Though the numerals were different, the order of synergism did not vary by both formulas. For similar concentrations used, $GI_{12.5}$ combinations of individual agents had least effect, while GI_{25} concentration combinations had the highest effect, which is self-explainable. Between the mixed combinations, increasing FXY-1 with low-dose I3M had more efficacy when compared with the vice versa combination.

Indirubin and its derivatives has been proved to have broad-spectrum antitumorigenic activities in a variety of cancer cell lines which includes lung cancer cells, breast cancer cells, prostate cancer cell line, translational cancer cell, murine leukemic cells, renal cell carcinoma, and human myeloid leukemia, (Gaboriaud-Kolar et al., 2016; Schwaiberger, Heiss, Cabaravdic, Oberan, & Zaujec, 2010; Wan-Yu & Nai-Wen, 2013). I3M has been demonstrated for diminution of cancer cell proliferation by inhibiting the ATP-binding site of cyclin-dependent kinase (CDK) (19). Another novel derivative of indirubin 3-Oxime, AGM-130 has been proven to inhibit the proliferation of A549-lung cancer cell proliferation by arresting them in the G_2/M phase of cell cycle, ending up causing apoptosis by altering the Bcl-2, Bax levels (Ahn et al., 2015). Both in silico docking studies and biological evaluations have shown FXY-1 to inhibit lung

cancer cell proliferation by binding to the PH domain of AKT enzyme (Rajagopalan et al., 2016). It is because of the high degree of overlap between different signaling pathways within the cells, agents with multiple mechanisms of action are particularly useful in combination treatments. The results of present study demonstrate the synergistic efficacy to be bought out by the two different mechanism of action together in lung cancer cells. Reports indicate that uncontrolled proliferation of the cancer cells may lead to cell invasion and migration, finally ending up in metastasis (Zhang, Lin, Ma, & Zhao, 2019). Control of these three mechanisms will effectively inhibit the progression for any cancer (Zhang et al., 2019). We mimicked metastasis of A549 and NCI-H 460 cells by cell scratch-wound and transendothelial migration assays in vitro. The observed efficacy of the individual and combination treatments clearly demonstrates their antimetastatic properties. During the metastasis process, cancer cells are reported to undergo transition change from epithelial to the mesenchymal form, which is described as EMT which is supported by biochemical and physiological alterations like cytokine expressions in the cellular microenvironment (Hsiao et al., 2016; Shang, Cai, & Fan, 2013). When stimulated, these cancer cells release $TNF-\alpha$, to support EMT process (Rhee, Lee, & Eom, 2015). The inhibitory effects of I3M + FXY-1 for $TNF-\alpha$ release in A549 and NCI-H460 cells further augments the observed antimetastasis effects by these combinations.

To further evaluate the observed antiproliferative effects, we checked the cell cycle status of I3M, FXY-1, and their combinations

in A548 and NCI-H 460 cells. Cell cycle regulation remains as an important event to decide between survival and death particularly in treatment regimen of cancer cells. One of the hallmarks of apoptosis is the digestion of genomic DNA by endonuclease leaving the cells with reduced DNA content (Yoshihiro, 2004). The present results also demonstrate that I3M, FXY-1 or their combinations inhibit the growth of A549 and NCI-H460 cells by inducing DNA damage as revealed by characteristic changes in M4 or Sub G₀ peak of the flow cytometry histogram. Reports suggest anticancer compounds could initially arrest the proliferating cells in the growth phase of the cell cycle and then induce cell death by apoptosis (Perabo, Landwehrs, Frossler, Schmidt, & Mueller, 2016). Our observations of the reduction in the G₁ (M₁) growth phase and subsequent increase in the sub G₀ (M₄) phase of individual and combination treatments stands in line with these observations.

Apoptotic induction in cancer cells is another approach to limit their uncontrolled proliferation (Kaur et al., 2009). Therefore, any compound that has a potential to induce apoptosis can be considered as novel drug for cancer treatment (Mohamed, Hidemichi, Ali, Yusuke, & Noriaki, 2014). One of the early events in apoptotic process is the translocation of membrane phosphatidyl serine from the inner side of the plasma membrane to the outer surface. As the membrane integrity is an important phenomenon for a cell undergoing apoptosis, Annexin V binds to the phosphatidyl serine moiety of any apoptotic cells and to differentiate the early and late phases of apoptosis (van Engeland, Ramaekers, Schutte, & Reutelingsperger, 1996). The occurrence of dose dependence early, late, and total apoptosis in both lung cancer cells (Figure 5) with individual and combination treatments confirmed the apoptotic events in support to the synergistic effects observed in proliferative assay.

Apoptosis is generally controlled by a cluster of gene products that either endorse or suppress the cell death during different stages of the cell cycle. Bcl-2 gene family is one of them that blocks the release of cytochrome C out of the mitochondrial membrane and thereby prevents apoptosis (Choi et al., 2011). In contrast, the Bax protein family interrelates with outer membrane proteins of the mitochondrion to release cytochrome C, thereby triggering apoptosis (Choi et al., 2011). A ratio of Bcl-2/Bax has been shown to be crucial in determining the susceptibility of ant cell that undergo apoptosis (Rasiova, 2002). From Figure 6, it is evident that down regulation of antiapoptotic Bcl-2 caused an increase in the pro-apoptotic Bcl-2/Bax ratio in both individual and combination treatments, suggesting that Bcl-2 participated in the apoptosis of lung cancer cells. These findings clearly demonstrate that the combinations induced apoptosis in both the lung cancer cell types, which could be accounted for the observed antiproliferative effects. As the low ratio of Bcl-2/Bax appears to increase chemo sensitivity, our data suggest that the combination treatments can be benefited for therapeutic abilities against cancer. Moreover, it is noteworthy that, the efficacy of the combination remained very close to each other in most of the assays for both the cell lines. It can be postulated that, since the concentrations of the individual drugs in the combinations were selected based on their GI₅₀ values, through the numbers were different, the overall effect observed in combinations remained close for both cell

lines. Therefore, these observations provide promising strategy for effective formulations for future experiments in treating lung cancer.

In recent times, advances in cancer research has increased the scientific knowledge of carcinogenesis at the tissues/molecular level, where target-based anticancer agents are designed to challenge the tumorigenic process. Therefore, using interactive complex combinations have been proven in terms of efficacy and safety against cancer types (Khafif, Schantz, Chou, Edelstein, & Sacks, 1994). Several studies indicate, less efficacious drugs, when used as combinations, exhibit synergistic activity in cancer cells (Suganuma et al., 1999). In line with these observations, our study demonstrates clear synergy in the lung cancer cells with I3M and FXY-1 at lower concentrations.

5 | CONCLUSION

In summary, I3M and FXY-1 showed anticancer effects in lung cancer cells exhibiting synergistic effects in their combinations. As the synergism is achieved at lower concentrations, it provides a promising approach for treating lung cancer with the lower dose levels without compromising on the efficacy of treatment. Further research is recommended to develop these results in transforming to successful combination therapy against lung cancer.

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CONFLICT OF INTEREST

The authors declared that they have no conflict of interest.

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